

ActInf Textbook Group ~ Cohort 5 ~ Session 22

(Chapter 10, Part 2) ~ 4/8/2024

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all right welcome back call 5 we're in our last recorded meeting Andrew and all of us will be talking about chapter 10 but also kind of probably anything goes and yeah let's go there Andrew however you want to proceed yep sure yeah I'll just give a quick kind of summary of this chap I cannot give an entire holistic summary of this chapter because as we know it longest chapter in the textbook um and one of its key themes is to describe active inference is a basically a kind of complete solution to sentient organisms adaptive problems which attempts to provide a unified perspective on action selection perception attention emotion regulation of right of other cognitive mechanisms um so it's it's it attempts to give us I think a kind of like let's round out the textbook here let's let's let's go over in section 10.2 all of the previous chapters up to this point give a quick summary like kind of in a per paragraph approach for each chapter reminding us of anything from marov blankets to neurobiological sub substans message passing uh building models and so on um section 10.3 quickly mentions this uh the famous uh philosopher Daniel dennet his suggestion to model the whole iguana uh was the was the quote and that's more or less what active inference attempts to do by using free energy minimization is a sort of first principle by which we can derive implications for a variety of cognitive processes um it takes a kind of inactive integrative approach to to behavior to preference realization self- evidencing self-preservation and autopoesis um it does not separate entirely action and perception as is done in uh or has been done at least in different um neuropsychological uh approaches and uh yeah it just attempts to yeah just take a very integrated view don't view action and perception as totally separate don't view Memory and attention is completely separate planning prospection retrospection all these things are are tied together by way of a free energy minimization and then from there for anyone who's you know more more focused upon sort of the the neural and neurobiological substrates and underpinnings neural signatures getting into neuroimaging Data then it might just be the case that we're looking at different Pathways uh for each of these things but nonetheless they're

they're sort of of a piece um and before I I wrap up this this summary I mentioned I I noticed at the end of section 10.3 they mentioned um uh reference this this great neuroscientist uh up at NYU Yori busak who's kind of famous for um back around 2006 I believe 2004 2006 um talked to began talking about this kind of inside out approach to studying the the brain and how the cognitive and psychological Sciences have sort of inherited all of these um Jamesy and William jamesian uh categories in in how they understand the the mind the the mind brain connection and so on so these were even these very words that we're using uh memory and planning and emotion uh you can just flip open like one of James's like key books that's been used as a foundation in many Psychology and Neuroscience courses and just yeah look at its table of contents and we have so many of these these words that that are continuing to be used in a very sometimes very free floating kind of way uh in in current research um what does memory mean uh what what is attention exactly and uh yeah so it's it's not to say that all these jamesian categories should be you know eliminated uh from The Sciences or you know reject did or even uh supposed to have you know zero value they have plenty of horis value but uh it's just it's an interesting take to try and use what we know now about neurobiology about computation mathematics and yeah just maybe it's time for the the cognitive sciences and others to kind of update some of their their taxonomies and terminology uh that they use um for this kind of work and and just consider like more integrative perspectives and for all these things that are treated separately so anyway that's kind of my two cents on that reference they make and um so from there we get um you know these kind of mini subsections uh that relate active inference to just a broad variety of other fields um its relationships its different is um because they're able to find kind of positive relationships between active inference and these other fields it kind of opens the door for dialogue uh for control theory predictive processing um action control cybernetics idiom motor theory it's it's I think it's it's kind of it's it's doing exactly what it purports to do which is to find an integrated perspective um then that said it distinguishes itself um active inference is becoming its own kind of approach framework uh body of knowledge and so yeah it's uh it's just kind of placing itself in the sort of epistemic field that we we find ourselves within and um relationships with utility decision making reinforcement learning so many people I think come to active inference in part because they want to study those things whenever they're not coming from a more neuroscientific background and it's like well utility is kind of implicated in active inference but but it's more than that we have epistemic value and we have balancing between epistemic and pragmatic value rather than treating them as their own separate cost functions or ignoring

one of them or otherwise and finally the the chapter points at sort of these other fields in open directions um the idea of modeling social and cultural Dynamics is a kind of broader scale application of active inference um you could even presume that that goes down to the micro level that is we have human beings individuals with their own generative models just interacting in a kind of multi-agent environment and then from there checking to see what are the emergent behaviors whenever they interact and so what kind of kind of social dynamics happen what kind of cultural creations and Productions they they come up with um things like that so um I think I'll leave it there for now there's just too much to go over I won't give like a point by Point like here's the difference between active inference and um you know reinforcement learning right away or something but um that's about good stopping point I think so if anyone has any you know points of interests they want to talk about any kind of questions something on their mind U please feel free to go for it okay I have a have a more of a it's yeah try to explain it in simple words like I've you seen this use of this word epistemic a lot and it can make multiple things what does the word epistemic mean in this in the context of active inference yeah sure I'll I'll I'll give that a shot um so previously in the textbook at times we would see you know we've seen different like kind of equations that break down computation of variational and expected free energies and so we've seen how it can be broken down into word like certain terms that that have been given a nam so we have like Risk ambiguity um complexity accuracy so so two of those terms that can be broken down into are pragmatic um value and epistemic value and uh those are those are pretty essential they relate to uh a classic problem in the the social sciences called the the explore exploit tradeoff um but in simpler terms pragmatic value as opposed to EP epistemic is something like an agent just going for realizing their goals directly like you're hungry you want some food you go for the food um you know realize that preference to to you know diminish your hunger keep your homeostasis pretty aligned and balanced um but epistemic value meanwhile would be acting in order to get more information in order to realize your goal so again you're hungry you want some food well where can I get food like maybe it's contextual where can I get food right now um I you can use the internet search up what places are open uh what what places are delivering um how much money do I have uh currently and what's my budget so it would basically just be kind of instrumentally you searching seeking for information that can help you to realize the goal so often times Whenever there are like behavioral experiments like a kind of two Arm Bandit problem for example where an agent can uh they can go to uh they have multiple actions available to them they can go uh immediately to uh a reward like pulling

the slot machine at the the end of the task or prior to that they can go to a queue that will inform them uh Which slot machine to pull which one has the higher probability of giving them a payoff and so the agent um you know if the if the agent doesn't care about the epistemic value it could derive from the queue uh giving it the information of how to better realize its goal it'll just skip the Queue go straight for pragmatic value pull the slot machine see if it wins um so I I I hope that that was helpful but I yeah I think that the simplest terms are epistemic is something like the knowing it's the it's the information that'll help you get to your goal pragmatic action and value are derived from from actually going directly at the goal so you collect enough info you know what to do after that you go for the pragmatic value yeah a little bit of just other broader pieces here epistemic in contrast with belief doxa like a Creed also as distinguishing the the knowledge from the Creed from the technique which is the basis of Technology um and then another helpful piece like uh Andrew mentioned the epistemic pragmatic value the epistemic value is like known unknowns in the prospective setting because it's a value ascertainment about a given policy how much I expect to learn from that policy so of course if something you didn't expect to learn happened it couldn't have been considered at the moment so that would be an epistemic event like in the broader sense of just epistemic meaning learning or knowledge but when we're talking about expected Ed free energy equation 2.6 epistemic value is specifically about expected Information Gain which has to do with like how certain or uncertain the agent is in the moment and then the epistemic chaining demo um which is just like one step it's basically One Step more detail than in the textbook but um you see the mouse navigate to The Q but this just kind of begets other questions like how does it come to believe with with full Precision that the Q is informative so this isn't the whole iguana but this kind of isolates a situation where either a risky pragmatic decision can be approached immediately or a little bit more of a circuitous path can be taken that results in Information Gain results in observations that are expected to reduce uncertainty about where the preferred observations are and then depending on how all the parameters are set you could have an agent that just still goes and risk it risks it for the uncertain reward sooner can we can we actually phrase it like uh epistemic uh value and epistemic capacity is like the potential of the possible things which can be done and the pragmatic things the actual thing that we do so it's like EP epistemic values are like uh as a ground on which possible things can happen and pragmatic thing is the actual thing that you do like conversationally that might make sense but when you're talking when you say things that you could do that's basically affordances and affordances then can be evaluated in terms of their epistemic and pragmatic value but it isn't

that something that you can do is related to The Information Gain of that action but I mean conversationally it makes sense like the broader epistemic concept the agent has to know what it could do yeah it has to be known what could be done so it's not that other terms are expunged of epistemic components it's just that epistemic value is being used in a pretty narrow way in equation 2.6 in the expected Information Gain that does highlight something very important but you could do kind of an epistemic inquiry into how does one know that they know and so on sorry I just uh could you just clearly differenti between this with affordances which you said I think what I saying is like so affordances is the possible things which a person can do at a particular moment of time right exactly yeah and policies like the the various policies you can do but affordance is like at that specific point of time what the agent does that's basically the affordances so affordance is very context sensitive yeah affordances are actions that can be taken in a moment policies are affordances over the time Horizon of planning so if the time Horizon of planning is just one then the policy space is the affordance space if you're talking about a time Horizon of two the policy space P_i the list or vector P_i that sums to one if you're talking about a time Horizon of two is going to be the length of affordances squared so like up down left right in the moment affordances and then planning Horizon of two so the policy space of which consideration is happening in would be 16 and then each of those 16 is going to have a prior so How likely is it a priori and then if it gets updated by expected free energy it's going to get updated according to the Information Gain and the pragmatic value of each of those policies where did you get the number 16 like four into four four squared oh okay yeah and that's the kind of just like it's like a chess algorithm like the first pass approach is just to expand your policy space exponentially but then if you have more than a couple of affordances at each time S_t or you go more than a couple of time steps deep the space of the policy Vector becomes huge and then that necessitates all these different approaches to dealing with large policy spaces ranging from like pruning of policy trees to setting up the problem a little bit differently so that the branching is less exponential so here's the narrow epistemic value concept just KL Divergence fancy e big expectation of KL Divergence between r q variational approximation of hidden States Through Time conditioned upon in both besides the action everything is conditioned upon the action and the only difference is why Tilda so for that action how much will those observations coming in update my beliefs about hidden States through time if the observations are expected to be moot then the Information Gain is low because it's like it might as well not be here if you expect there to be an observation coming in that's super informative then this is

going to be a large Divergence and that's kind of perfectly partitioned away from whether those observations coming in are aligned with what you expect and prefer to see like pulling a slot machine like I mean there's there's a lot of ways to go with this but like you can imagine there's two agents they're both pulling a slot machine one of them is doing it picking that policy because they're seeking the information that the slot machine provides next another one is doing it because they believe that it's going to come up in a favorable setting for them so that's kind of something people people are exploring with a Precision Psychiatry would be okay this person's engaged in this repetitive behavior is it because it's simply a habit and they don't prefer to do it is this happening because they actually prefer the outcome that it reaches is this happening because they prefer an outcome and they've misunderstood the consequences of the action so they think that that action is going to reach their desired outcome but those from the outside look exactly the same hence designing like experiments and cognitive models that could differentiate why those are different I have one question could you please go back to that last equation which is showing yeah over here actually I don't know what this what does the term mean this Q the last one that that what is the operator equal to and a triangle on top any idea defined as equal to okay defined as equal by definition two equal by definition to like this is the joint distribution so here's Q that's our variational approximator that we control this is a joint distribution of X and Y conditioned upon P_i so it's like there's kind of an X and A Y field and we're making a probability density over the X and the Y field conditioned upon what action we're selecting and then that is defined as the joint product of our beliefs about hidden State given policy multiplied by basically the A matrix the real mapping between hidden States and observations so here's the here's our our big underlying joint generative model of the true temperature in the room and the thermometer reading conditioned upon policy and we're going to be able to break that down into well what is the temperature of the room given our policy and then what is the thermometer reading given the true temperature of room because of the factorizability the the fact that the X and the Y were orthogonal gives us a sparsity that we can leverage to decompose The Joint distribution into these two simpler distributions e that's quite a thread there it's like I've forgotten more than I'll ever know just like how did this happen deny the meaning of awareness that's um I do recommend I cannot remember the name of the paper but there's like this classic Marvin Minsky paper where he's describing his take on AI at the time this must be decades ago at this point and at the end he hints at the idea that uh and some kind of artificial intelligence would have to have a model itself and I just remember skimming that paper and looking at active inference and this idea

that one has to have their own model of themselves in the world in order to kind of maintain that that Markov blanket maintain that separation between self and others so that you can actually maintain your own homeostasis maintain that um division between between model an environment kind of rambling on yeah and like like we're aware of making decisions on top of what we're not aware of we're able to talk about this semantics of like moving our hand around because there's this kind of infrastructural component that we're not generally aware of they call it self-evidencing right yeah well self-evidencing let's see where it comes up self-evidencing is provided as something like the analogy to reward maximizing which puts reward maximization as the unified imperative in active inference where we have surprise minimization or surprise bounding about one's generative model as the kind of key component we can think about as I say one existential imperative self-evidencing but solipsism is kind of a um deadend strategy so self-evidencing doesn't just mean observing oneself because over long enough timelines that ends up becoming maladaptive so that's kind of like well how could novelty or creativity or Aesthetics be brought in when the unifying principle that underlies behavior is to glean evidence for one's model as it is this is funny also like in chapter 10 has to come back to the preface so this is I guess written by Carl it's not just about reading the free energy principle equates existence with self-evidencing can I tell a joke like I don't know it's a joke or something they they said that like all these people this they're just worried about it the mind will just shut off you know like they will not do any self-evidencing and then uh no nobody wants do that something that is like Mo like like the greatest fear among many people is that okay they just don't want them they're just worried about that the mind will shut off one day and there'll be no self-evidencing and everything will go at us you know all this kind of stuff h sorry my level of jokes are with my kids like the way I speak with my kids that is the level of jokes I I don't know better ones that's the sweet spot for humor though it's like three to six years yeah I mean it's it's a strongly reinforced theme that simple on through complex systems are self-evidencing to Garner sensory data consistent with an internal model so it's like in the reward World we'd have some temperature distribution and then we have a second layer Distribution on top of temperature how rewarding those temperatures are and then we pursue rewarding temperatures like body temperature in the self-evidencing world we have our basal distribution of temperature and then it's just explained as action is taken to continually sample from that distribution of preferred temperatures whereas reward you need to interpose with like another layer in between the temperature and the action so you're you have temperature and then reward layer and then you're taking actions based

upon reward of different actions here we have the basal temperature layer and then we're taking actions directly on that distribution um can I give a perspective of self-evidencing as it relates to meditation yeah yeah okay I'm I'm not sure if I'm in the right track like there will be more learnings but self-evidencing like it's actually people self-evidence is there is a there's actually opposing force between self-evidencing and meditation basically because self-evidencing is you want to that's how I'm thinking like we know ourselves by the evidences which are provided from outside that is we are inferring our evidences from the knowledge from outside and um meditation is trying to see what is really inside so and and then there is one more this concept marvelous concept of blankets right so it's conditional Independence you if the more you know about the outside world if you know the blanket States doesn't mean that you know more about the inside world and that's like a that's that's like I I I have just fallen in love with this fundamental concept like even if you know so much about the external world but if the blanket states are just this much then knowing more about the external world doesn't know more about yourself so self-evidencing has its limits and uh yeah so there is um there is some kind a dynamic interplace so but but what I think is this this this this formulations and this concepts of active influence can let us know so much about the mind and how it works uh but there will be there I don't know much but like there will be challenges when we explore the Realms beyond the mind it's the Consciousness so I don't want to talk about that because people can say and they can have lot of theories but it may not necessarily be right because this conditional Bound by this Marco BL blanket presents a very fundamental um that's how that's like I don't know if I phrased the appropriately but like that is like this these two pillars are what I'm basically thinking and how it relates to meditation and these Concepts yeah yeah it makes me think of U I don't know if this directly relates to what you're describing but this whole you know just because you might know a lot about the outside doesn't necessarily mean you know a lot about the inside and it makes me think about you know similar to the rest of of kind of current research and n neuroscience uh active inference also splits up uh when describing different sensory modalities as you know we don't just have like one kind we have proprioception and exteroception and interoception and all three of them relate to inference and kind of having to you know update beliefs change your attention um and and and kind of figure out what to do and so in Tarot SE means like reading your own body and reading your own feelings even this this chapter 10 here relates um there's like a section entire section that relates to like emotion um in in trying to describe that right like you have to actually infer how you're feeling you don't just automatically you know perfectly

precisely know and then suddenly an emotion isn't so much like some kind of innate perfectly inbuilt thing that we just automatically like understand like oh I'm sad it's like no you have to infer that you have to like read what's going on inside of you just as much as you have to read what's going on outside of you so suddenly your emotion is yeah it's like a hypothesis of like oh based on this feeling and this and this and this other sensory information that I'm getting from within me like how am I feeling how am I doing let's put a conceptual you know word on that name of some emotion um yeah interesting uh Kristoff you have a your hand up uh yeah I just posted a link uh to a talk by uh Shamil uh chandaria I think he's affiliated with University of Oxford and a bunch of other places and it's a um Nono all Awakening yeah computational neurom phenomenological account uh which is basically entirely based on the free energy principle active in and it's an it's an incredible talk it's the best summary of um essentially various aspects of spiritual practice from the perspective of computational Neuroscience that they have ever encountered and it's is really really he goes really really deep um I can tell you like from from my perspective um understanding the free energy principle um has been enlightening to to say the least like in in the in in the for me it's actually in the literal sense there's a concept in Buddhism which I believe is uh Liberation by view um where um you know a or a practitioner um receives a certain level of of of Liberation by essentially understanding um um certain truths about the nature of reality without necessarily realizing um uh for themselves a a certain meditative attainments which are um available to practitioners in the path of insight so if you practice mindfulness and concentration there is a whole ladder of um uh insights that you you essentially climb you know the the the four janas the the for the formless Realms and finally the attainment of arand ship and I find it that when I look at the teachings of the Buddha particularly um you know say reading the nikayas the maim Manaya the middle length discourses or s n the connected discourses and thinking about them in terms of um free energy principle and active inference um the teachings of the Buddha are incredibly clear like he is giving you recognition models he's giving you means of building your own um uh generative models and it has a lot to do with essentially making the pH phenomena that happen in the mind opaque because normally they're transparent to us we don't know how they work we don't know how the reality is uh constructed by the mind we are not aware um of causes and effects that give rise to certain thoughts and feelings and and perceptions and and mental formations and concoctions and all these other things and Buddha is giving you essentially a vocabulary so that you can um later on label these things as they occur so he's giving you a recognition model you can say like ah this is a this is a perception

this is a mental formation this is a feeling and feeling is born of contact and contact is you know between the internal sense basis and the external sense basis and so on so forth and then it that's for the purpose of effectively building a generative model um once you have the generative model then you know you you get you become you can become more introspective about your own experience and um the the goal of that is that your experience becomes increasingly opaque uh you see more aspects of it and so on so forth and uh at the end of the path you basically have complete insight into how the reality is constructed by the mind and that's why this phrase always um recurs in in the Buddhist text that you know um um the biku who completed the path understands the the the birth is destroyed the holy life has been lived what what what had to be done has been done and there is no more coming into any state of being it's not a it's not a matter of you know question like once once you get there you know you know you're there so you know like it's um I I think I I think there is a the you know going back to this earlier statement about the necessity of updating the terminology and taxonomy once you start looking at at Buddhist teachings through this perspective and if you look at this for this modern perspective of computational Neuroscience it's it's an Incredible manual and it's so straightforward there is no there is no um there's no doubt about what you need to do and how how to do it so I highly recommend um this talk and uh you know it's a it's it's it's a very fruitful area of inquiry awesome Francis thanks uh firstly uh I'd endorse that uh thanks for um poting out to me Kristoff uh I uh watched it twice uh I'm sure I've still got a hell of a lot more to get out of it um it's a really interesting um secondly just kind of a related um things to a very simple but it struck me as interesting point uh which relates to the whole idea of making the transparent uh opaque um and the fact that by making it a p you can see it which of course sounds a little paradoxical um I've been thinking about um mindfulness and kind of phenomenology because it's really hard phenomenology you know what is it that we're feeling this stuff that we're not normally aware of and of course a common mindfulness technique is to label your thoughts and of course we normally think of that is oh we're labeling them because we want to then let them go but of course by labeling them you make them opaque and you could get a very good stream of Consciousness um uh by just speaking these labels loud um by pursuing the um St of mindfulness and the empty mind you actually become aware of what's in the mind and then um the third thing just kind of going back to this homeostasis interception um and something even simpler uh which is when I was first started studying to be counselor I was introduced to uh C's uh learning Loop uh Kolb and that has four axes which are it's kind of used in an educational context and uh they have these kind of rather long

labels but then there's another variation where the four axes have one word labels and the labels would be uh action feeling thought planning and that I've explained that kind of gives us a model of how we process our feelings um and I explain it to people in a counseling context why is it count counselors are always asking you how this makes you feel how you feel about this but you know we've got this engine this cognitive engine up here but how do we know what what's steering it what's the point our feelings are where the rubber hits the road the road being our well-being if we don't know how we're feeling we don't know what to do with our cognition um and as you get further away from basic biological homeostasis homeostatic demands the biological needs uh it gets harder and harder to engage with are more complex in a leads demands between say status and connection uh should I pursue a job or should I make time for being with my partner uh and especially when you enter this uh existential anomaly all these brain power cognitive power helps to increase our survival that's what it's there for but what happens when we get so much cognitive power that we realize that none of the all all our paths lead to nons survival in the end and into the existential challenge little dump mind dump there can I just add something about labeling um popped in my mind um as you were speaking uh so that's there's a that's mahasa tradition um uh is essentially to just practice labeling all day long you just label everything you do but um I found I found I find this uh really there's this really interesting paper um from the deep Learning Community called your classifier is secretly an energy based model and you should treat it as such and um I think it's by Will Grall grad wolf I'm not sure what the C name is but anyway so they show that if you train um you know standard classifier train on across entropy objective like typical you know amidst class ifier neural network there's a way of interpreting that as a energy based model in such a way that you get effectively the kind of unnormalized probabilities that are required to compute free energy and you effectively get um uh a generative model for free so this idea you know of labeling by m and and other traditions in Buddhism that all you have to do is labeling it's actually uh you know if you squint your eyes at this paper and you say like you know learning is kind of universal and no matter how it's implemented the the same sort of effects take place you are B that that that is exactly the way to build a generative model um which I find I find it I I think this this formal equivalence between uh classification uh uh you know the recognition and generative models through energy based models specifically which is just such a beautiful enrich um formalism especially and and so connected to to active inference um that's going to be um is going to be important so I I think this is a very uh very interesting uh research Avenue of how to connect active inference with um this deep learning world and

in some sense you know like the because active inference and free energy are so Universal phenomena we know through this argument that because they're apply to anything that exists mathematic a prior then these these systems must be building models in the basian sense from a certain perspective but it seems that that equivalent is a lot deeper and much more direct than just um you know this kind of this is a mathematical principle that has to be true kind of argument it's actually can be shown how uh how that happens so yeah uh fascinating paper I highly recommend it yeah super cool it's fascinating to see just like with this the integrative approach of this last chapter how many possible topics we could land on during this conversation from from Deep learning mathematical formalisms uh the excesses of the the human cognitive mind um yeah it's it's it's really interesting I personally just I find this kind of first principles approach of act active inference to just open doors to many different dialogues um and Francis I was finding that to be very interesting I had like a a kind of naive take on uh the idea of like what what happens whenever you start getting away from like a parent homeo static needs um from an actm perspective which is you know from a a modeling perspective we're looking at like an agent has their preferences uh which are really their their expectations uh about preferences and and so there are things that you expect more to occur um and those those are you know your your wants your needs and so on might be some kind of colloquial language we could assign to those uh prior or those preferences and like the idea might be that as a human being navigates the world going forward an increasingly more complex world you know they they start to learn that they need to do things Beyond simply oh I need to drink water to say sh thirst um like a variety of things navigating a kind of Legal Financial social occupational familial world that they're in and it's almost like there's this kind of expansion of uh of the state space to learn about and uh with that a kind of expansion of observations or preferences to be realized in order to to meet our wants and our our needs and and and so to bring it back to this discrepancy between like oh just keeping your heartbeat uh regulated keeping your your hydration levels reasonable like suddenly there're all kinds of things I want a nice house and I want a I want a nice car or or conversely I I want to have a very energy efficient uh green uh lifestyle uh or some such thing and and as it becomes more complex it's like it just makes me think of uh this classic Rolling Stones song uh You Can't Always Get What You Want um and you know sometimes you can get what you need um it also brings me back to Buddhism and a lot of commentary upon the role of of desire and its relation with with suffering um just kind of you know Li desire can ultimately be be Limitless it can be transient uh it can be something that sort of leads you astray from the path um can be all kinds of things

and yeah I don't know I I don't know if it's about kind of raining in our our desires every once in a while and really evaluating like what is important here uh what do we really need to do um and and how can we how can we have our basic needs met then from there I look at like look to neuroscience and uh folks like um there's someone named phel uh Sarah Felman Barrett she's great she's done a lot of work on U uh emotion and affect and and she's at least a couple of times had some collaboration with Carl Friston and other folks in active inference um and she says like sometimes you just need some good sleep some sometimes whenever there's a kind of overwhelmingness a kind of you begin to interpret what's going on with you is like anxiety uh you know in the extremes of kind of catastrophic thinking of trying to do everything all at once you can imagine an active inference agent trying to just minimize a ton of free energy all at one time and it cannot find the right policy to to accomplish that within five seconds as much as it might wish that such a thing were possible um yeah it's really just you know get some good sleep can you do that and you know get some get some good nutrients in your body and maybe get whatever kind of exercise is reasonable to you if if only like a nice walk um apparent that can heavily change the the inferential process uh and maybe can help with straightening out thinking um I could ramble on forever but yeah I'm just kind of uh wrapping up the the meeting here some thoughts throwing them all around I have to leave in one minute because I've got a client uh on the hour can I just uh yes please very thank you so much and there's so much more love to say and to talk about uh and I'm really sorry that we've come to our last session yes for this little fractal cycle but we will all be able to suffer more looking at pdps in the future I'll be there with luck okay thanks guys well thank you so much thanks everyone yeah thank you okay next week um we'll we'll just see what happens we can um fill out the feedback form anonymously look through some of these many comments and also I think now that we are two years after the the publishing of this book March uh 22 we can start to think about you know we have RX and fur we have all these other activities how do we want to continue with the par our textbook group so think about it over the next week and then we'll discuss more L later thanks thanks Andrew thanks everybody thanks everybody it's been amazing yay thanks folks